

Kaushic Aravind B

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EXPERIENCE

ML Engineer | Scimplify

Jun 2025 - Present | Bangalore, IN

- Collaborated directly with the founding team of an early-stage chemistry AI startup, including engineers from **OpenAI, Google DeepMind and Sarvam**, to design **evals and a benchmark dataset** for empirically measuring agent performance on chemistry research tasks.
- Designed and shipped **Scira**, a production **LangGraph research-agent ecosystem** used by **300 scientists**, owning workflow design, agent orchestration and deployment; integrated domain-specific foundation models and reduced research turnaround from **5 days to 1 hour**.
- Designed the agent runtime for **secure multi-tenant execution** using Landlock-sandboxed tools, row-level authorization and **LangSmith LLM-as-judge evaluation gates**.
- Re-architected **OpenSearch** with hybrid BM25/vector retrieval, metadata filtering and cross-encoder reranking; improved **recall by 30 points** and reduced **retrieval p90 latency by 80%** at **10K queries/day**.
- Fine-tuned **Gemma 2 27B with SFT/LoRA for structured output** in the same search workflow, reducing **model p50 latency from 15s to 3s**; used **Laya for simpler NER** and **Redis LangCache** to avoid redundant inference.
- Built a chemical knowledge graph with entity resolution, deduplication and write-time validation; engineered a **PySpark pipeline loading ~1 TB into BigQuery in 2 hours**.
- Trained a **Transformer + MCTS retrosynthesis system**, achieving **88% top-1** and reducing chemical-design cycles from **4 days to 15 minutes**.

Data Scientist | Sulekha

Sep 2023 - Jun 2025 | Chennai, IN

- Built and deployed **QueryLM**, a fine-tuned text-to-SQL service answering business questions without exposing database schemas; reduced **RAG cost by 60%** through rechunking, caching and pipeline redesign.
- Built a production **hotel-booking agent** integrating reservation workflows and a payment gateway; reduced **inbound calls by 80%**.
- Architected a **Gemini audio intelligence platform** processing **100K+ agent-hours/month**.
- Built a multi-task **POS-BERT-BLSTM-CRF opinion-mining system**; automated categorization and routing, reducing **grievance resolution time by 30%** and increasing **repeat visits by 40%**.

Applied AI Practitioner | Searce Inc

Jul 2022 - Sep 2023 | Bangalore, IN

- Led solution architecture across **4+ enterprise GCP engagements**.
- Built **document-intelligence RAG on Pinecone** with retrieval and response-quality evaluation.
- Delivered a **real-time speech-recognition pipeline for ISRO**, the Indian Space Research Organisation, with **Google Speech-to-Text and a live dashboard**.
- Migrated **10+ years of HubSpot candidate data** into a structured analytics store with resume parsing and hybrid retrieval.

INDEPENDENT PROJECT

Local-first personal agent | Independent build

2026

- Built a laptop-based **personal agent with Claude Agent SDK**, using a headed Chromium browser that supports user observation and takeover.
- Designed **persistent git-tracked markdown memory with offline consolidation**, retaining context across sessions.
- Implemented **vault-style credential handling through macOS Keychain** so secrets stay outside model context; protected sensitive actions with **deterministic, action-specific one-time-code gates** and **95 offline tests** across memory, safety, vault and operations.

TECHNICAL SKILLS

Agents & evals: LangGraph, LangChain, LangSmith, MCP, Claude Agent SDK, LLM-as-judge, benchmark design, A/B experimentation, agent memory, sandboxing

Programming & backend: Python, SQL, FastAPI, PostgreSQL, MongoDB, Docker, AWS, GCP

AI & retrieval: OpenSearch, Pinecone, Milvus, pgvector, BM25/vector retrieval, reranking, PyTorch, Hugging Face, SFT, LoRA, QLoRA

Systems & data: Redis LangCache, PySpark, BigQuery, knowledge graphs, entity resolution, multi-tenant authorization

EDUCATION & CERTIFICATIONS

B.Tech, Information Technology | College of Engineering Guindy, Anna University, Chennai | 2018 - 2022

Google Cloud Professional Machine Learning Engineer (2023) | Google Cloud Associate Cloud Engineer (2023)